

Threat Model

Grounded in: Dexcom G7 — connected mobile iCGM (cleared 2022)

Basis: STRIDE; FDA 2025 guidance structure; device facts public.

Device	Dexcom G7 — integrated CGM (iCGM)
Connectivity	Bluetooth Low Energy → G7 app (iOS/Android) & Apple Watch; Dexcom Clarity cloud + share
Method	STRIDE over sensor, mobile app, BLE link, cloud, followers
Note	Illustrative — shows expected structure for a connected mobile SaMD; not Dexcom's confidential file.

1. Decomposition & trust boundaries

- Entities: sensor/transmitter, mobile app, BLE link, Dexcom cloud, follower/share, Apple Watch.
- Trust boundaries: sensor↔phone (BLE), phone↔cloud (TLS), cloud↔followers.

2. Threats & mitigations (STRIDE)

Threat	Example	Mitigation	V&V
Spoofing	Rogue BLE peer	Bonded pairing, session keys	BLE security tests
Tampering	Altered readings/alerts	Integrity checks, range validation	Fuzz + unit tests
Info disclosure	Glucose/PHI exposure	TLS 1.2+, Keychain/Keystore	MobSF, capture
DoS	Alert suppression	On-device alerting independent of cloud	Fault injection
Elevation	Cloud/API abuse	Least-privilege IAM, scoped tokens	IAM review

3. Residual risk

RF jamming of the BLE link accepted; mitigated by on-device alerting and gap handling, disclosed in labeling.

Sources (public)

- FDA / Dexcom — Dexcom G7 510(k) clearance (Dec 2022); G6 first interoperable iCGM (2018, special controls).
- Public connectivity facts: Bluetooth Low Energy to the G7 app (iOS/Android) and Apple Watch; Dexcom Clarity cloud.
- FDA — Cybersecurity in Medical Devices premarket guidance (2025); OWASP MASVS for mobile.